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RESEARCH PROJECT REPORT

Final Research Report, Part IV Project #133

Making Education More Accessible & Reliable: AI-Driven Facilitator-Based Teaching

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ABSTRACT

Many schools cannot find or keep qualified teachers, and the quality of teaching varies from one classroom to the next. This project asks how a software platform can be designed so that any approved facilitator, regardless of subject expertise, delivers an expert-designed lesson as it was intended. The answer built here is StudyBuddy, a web platform with two roles: admins, who have teaching expertise, build lessons from nine types of block, and facilitators, who are present in the classroom, deliver them. Access is granted only after an admin approves a request, each lesson is walked through once with its steps unlocking in order, and every action is checked on the server. An AI worksheet generator drafts supplementary material that an admin approves at two stages. The platform holds more than 110 lessons across the NCERT curriculum (Grades 1 to 4) and the New Zealand Curriculum (Years 1 to 4). It was evaluated through a benchmark against existing platforms of three kinds on six criteria drawn from the literature, a heuristic review, informal design feedback, technical tests, and tests of authoring effort and the worksheet generator. StudyBuddy rates strong on decoupled delivery, structured guidance and adaptability, partial on accessibility and engagement, and limited on personalisation, because every student in a class sees the same lesson. In the technical tests, all 20 protected routes turned away signed-out visitors, all 19 forged admin actions were rejected, and the lesson-flow rules held in 13 of 14 checks. A lesson with prepared content takes about 7 minutes to build. These results show that the platform implements its design. They do not show that facilitators without subject expertise deliver lessons well, which needs a classroom study with real facilitators.

Signed Statement of Contribution

Student

I hereby declare that:

1. This report is the result of the final year project work carried out by my project partner, May Thazin, and I under the guidance of our supervisor, Professor Gill Dobbie, in the 2026 academic year at the Department of Electrical, Computer, and Software Engineering, Faculty of Engineering and Design, University of Auckland.
2. This report is not the outcome of work done previously.
3. This report is not the outcome of work done in collaboration, except that with a potential project sponsor (if any) as stated in the text.
4. This report is not the same as any report, thesis, conference article or journal paper, or any other publication or unpublished work in any format.

This is not a continuing project. In the case of a continuing project, please state clearly what has been developed during the project and what was available from previous year(s):

Nothing in this report was available from previous years. The platform, its more than 110 lessons, the AI worksheet generator, the evaluation and the account of AI-assisted development were all produced during the 2026 project. The proposal, literature review and mid-year report written earlier in the project are rewritten and extended here.

Contribution to the joint project

This report is one of two produced for Part IV Project #133, *Making Education More Accessible & Reliable: AI-Driven Facilitator-Based Teaching*, carried out by Moksh Manglani and May Thazin under the supervision of Professor Gill Dobbie. After a shared start, the project was divided into two complementary halves. Each of us carried out our own literature review, research question, method, implementation, evaluation and conclusions, and wrote our own report. The poster and the research compendium are joint work.

Moksh Manglani (this report): the literature review on current modes of education and their limits; the design and build of the StudyBuddy platform (Next.js, Supabase and Vercel), including block-based lesson authoring, lessons for the NCERT and New Zealand curricula, a step-locked delivery flow for facilitators, an AI worksheet generator whose output an admin approves, and the access controls around them; and the evaluation, made up of a benchmark against existing platforms, heuristic reviews, informal design feedback, technical tests and tests of authoring effort and the worksheet generator. AI tools wrote much of the code, as described in Chapter 7. The design, specifications, testing and decisions are my own.

May Thazin (partner report, Thazin, 2026): the literature review on AI in K-12 education, reading comprehension and simulated learners; the design of the reading activity and its interaction signals; the evidence-linked rule engine and the facilitator dashboard; and the simulated-classroom evaluation, its analysis and figures.

Signature:  Date: 18 October 2026

Supervisor

I confirm that the project work undertaken by this student in the 2026 academic year is / is not (strikethrough as appropriate) part of a continuing project, components of which have been completed previously. Comments, if any:

Signature: Date:

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Glossary of Terms

Term	Meaning
Admin	A person with subject and teaching expertise who designs lessons and approves facilitators.
Facilitator	The person in the room with the students who delivers a lesson without needing subject expertise, such as a parent, a teaching assistant or a community volunteer.
Class	A grade or year level, such as NCERT Grade 1 or NZC Year 1, which contains subjects.
NCERT	The National Council of Educational Research and Training, whose curriculum is used in India.
NZC	The New Zealand Curriculum.
Decoupled delivery	The extent to which educational expertise can be separated from direct classroom instruction (Section 3.2).
Implementation fidelity	How closely a lesson is delivered as it was designed (Carroll et al., 2007).
Heuristic review	An evaluation of an interface against a set of general usability rules, here Nielsen's ten heuristics (Nielsen, 1994).
Block	One step in a lesson. There are nine types: video, workbook, activity, worksheet, reading, quiz, discussion, break and note.
Walkthrough	The first pass a facilitator makes through a lesson, in which each block unlocks only after the one before it is finished.
Server-side check	A check carried out on the server, where it cannot be skipped by changing what the browser sends.
Forced tool use	A way of asking an AI model to return its answer in a fixed structure, so it can be saved directly and does not have to be parsed from free text.

1. Introduction

Access to quality education remains one of the most pressing global challenges in social and economic development. Although education is widely recognised as a fundamental human right, many communities continue to experience limited access to reliable and effective learning opportunities. This issue is particularly severe in rural, remote and underserved settings, where shortages of qualified teachers, inconsistent educational delivery and infrastructural constraints reduce both access and learning quality.

This project addresses that challenge through StudyBuddy, a facilitator-based education platform that aims to make educational delivery more accessible and consistent. The central idea is to separate educational expertise from classroom delivery: high-quality teaching expertise does not necessarily need to reside in every classroom if it can instead be embedded in the lesson design, the instructional flow and the support used during delivery. Under this model, admins, who have teaching and subject expertise, design lessons on the platform from resources such as videos, worksheets and activities. Facilitators, who are physically present with the students, use the platform's step-by-step interface to follow a guided lesson sequence during class. A facilitator does not need subject expertise as they are present to assist students and to navigate the platform, the teaching is done via the platform through the lessons that were created by admins. This preserves structure and consistency while widening the pool of people who can take part in delivering education.

2. Project Plan

2.1 Motivation

This project has a personal meaning to me. I grew up in Jaipur, India, and loved subjects like maths and science, but my friends who lived in the same apartment building disliked them. The main difference between us was our teachers. My father taught me maths. He is a great teacher and the reason I am able to study engineering today. He was very patient with me and made learning extremely fun. We would sit on the bed, and he would tell me stories, give me tons of examples and answer every question I had. He made me love teaching as well, and I have since tutored students myself, which showed me how much a single teacher influences learning.

I studied at Dolphins High School, which was well known in Jaipur for providing the best facilities at the lowest price, making education very affordable. In my years there I saw good teachers leave for larger schools and higher pay, leaving me and my classmates unattended and having to start again with a stranger, which affected our education significantly, especially in our earlier years. I later learned from teachers and parents that this is common in affordable schools.

When I travelled around India, I saw many children who could not read, write or do simple maths. When I asked around, I learned that these children have big dreams and desperately want an education, but have no access to it, and that, I was told, there are many children like this all over the country.

Hearing this, I was deeply shaken and wanted to see if there was anything I could do to help. I started volunteering, making small donations and helping out wherever I could, but I was always on the hunt for a more concrete solution.

2.2 Problem Space

Several interconnected educational challenges frame the problem this project addresses. The first is a shortage of teachers. UNESCO (2024) estimates that 44 million additional primary and secondary teachers are needed worldwide by 2030, including 15 million in sub-Saharan Africa alone. The second is that learning outcomes remain poor. The World Bank and its partners

(World Bank, 2022) estimate that 70% of 10-year-olds in low- and middle-income countries cannot read and understand a simple text, up from 57% before the pandemic. Getting children into a classroom is therefore not enough, and what happens inside it depends heavily on the teacher, because teacher quality is strongly related to how much students learn (Darling-Hammond, 2000). The third is cost. Meeting national education targets is beyond the means of many countries: the Global Education Monitoring Report team estimates an average annual financing gap of US$97 billion across 79 low- and lower-middle-income countries (Global Education Monitoring Report Team, 2023). Establishing and running a traditional school is also expensive and administratively complex, which makes it difficult for underserved communities to create sustainable learning opportunities of their own.

To get a deeper understanding of the problem, I reached out to my high school's academic coordinator for a quick conversation. She has over 20 years of experience in education and oversees 35 teachers. Three themes were evident from our discussion. First, the quality of teaching depends heavily on the individual teacher. Of her 35 teachers, she told me, "I can find five, six teachers only who really work for the children." Second, when a teacher leaves, whether for better pay, relocation or personal reasons, the disruption affects the whole school. Other teachers absorb the workload, management must find a substitute, and younger students are affected most because they form strong bonds with their teachers and struggle with the sudden transition. In foundational classes such as kindergarten, for example, children cry and become very upset when their teacher leaves. Third, she emphasised that regardless of experience, no teacher should ever be sent into a classroom unprepared. They need a briefing, time to become familiar with the material and a clear lesson plan.

In a broader view, the literature and the conversation suggest that the problem is larger than the number of teachers. The quality and continuity of a child's education depend on individual people: whether a good teacher is available, whether they stay and whether they are prepared. Education that is reliable, accessible and consistent therefore needs more than additional teachers. It needs teachers who are committed to their students, and a way of delivering instruction that does not depend on any single person. This raises the question behind the project: can learning in a classroom be decoupled from the individual teacher, so that students can still learn without an expert teacher in the room? In other words, what if anyone could teach?

2.3 Project Split and Research Questions

This question led to the idea at the centre of the project: a facilitator-based model of education, in which a lesson is designed once by someone with teaching and subject expertise and delivered by a person regardless of their pedagogical background. Making such a model work depends on two relationships. Since the lesson must reach the classroom as the admin intended, our first relationship lies between the admin who designs the lesson and the facilitator who delivers it. In addition to the lesson being delivered as intended, the students must also understand what is being taught. This brings us to the second relationship, between the facilitator and the students. The facilitator is not an expert teacher, so for a model like this to work, they would need to know whether the students understand what is being taught, and also be given guidance on how to cater to the students and respond when a lesson has not landed.

After our initial Literature review the problem statement was revisited and narrowed, splitting "reducing dependency on expert teachers" into two parts. These 2 parts were pursued in parallel, reflected the 2 relationships and were undertaken separately, converging on the same underlying goal of making education more accessible and consistent (Table 1). I undertook Part 1 and that is what this report covers.

Table 1. The two sides of the project

	Part 1: The Platform (this report)	Part 2: Student Learning Insights (Thazin, 2026)
Research question	Can any approved facilitator, regardless of subject expertise, deliver an expert-designed lesson?	Can a software system convert learner interaction data into evidence-linked recommendations for facilitator decision-making?
Purpose	Create lessons and guide facilitators through delivery	Understand whether students followed the lesson, and respond
Relationship	Admin and facilitator	Facilitator and students
What it covers	System architecture, requirements elicitation, lesson authoring, guided lesson delivery and the surrounding build	Year 3 to 6 reading-comprehension quiz and research-based rules that flag learners who may need support, with the evidence shown to the facilitator
Student	Moksh Manglani	May Thazin

2.4 Research Aim and Objectives

The aim of this project is to design and develop a proof-of-concept facilitator-based education platform that enables approved facilitators, regardless of subject expertise, to deliver expert-designed lessons through structured digital guidance, supporting more accessible and consistent education.

The objectives of the project are to:

- investigate the educational problem space surrounding access, quality and teacher dependency.
- identify the limitations of current modes of educational delivery.
- design a platform model that separates lesson design from classroom delivery.
- develop an administrative interface for creating and managing structured lessons.
- develop a facilitator interface that supports step-by-step lesson delivery, with access controlled by admin approval.
- incorporate AI where it adds practical value to the platform, such as resource generation, and use AI tools to review and stress test the platform during development.
- demonstrate the feasibility of the model through a working proof of concept.
- evaluate the platform against the needs and gaps identified in the literature.

2.5 Scope

The scope of this project is limited to the design, implementation and evaluation of a proof-of-concept platform for facilitator-based lesson delivery. The project focuses on the structure and functionality required to support guided classroom instruction rather than on the full deployment of a complete educational business or institution. Figure 1 shows the early concept that set this scope: an admin side where lessons are created, organised and approved, a facilitator side where a lesson is opened and followed step by step, and content security applied across both.

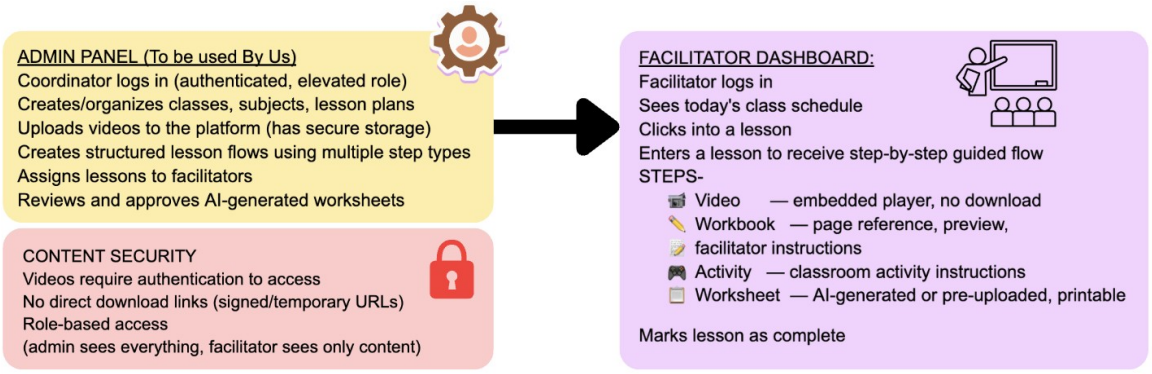


Figure 1. Early system concept used to set the scope of the platform

To scope the work clearly, it is also important to state what is outside it. The following are not covered by this report:

- large-scale field deployment;
- evaluation of educational impact or learning outcomes with students, since the platform is not being tested in a real classroom;
- complete payment, subscription or marketplace systems;
- full regulatory compliance workflows for operating schools;
- analysis of student learning and recommendations to facilitators, which is the subject of the partner report (Thazin, 2026).

3. Literature Review

3.1 Introduction to the Review

This literature review examines how well existing modes of education address the global shortage of quality education described in Chapter 2. Rather than treating education simply as school access or enrolment, this review is concerned with whether current educational models are capable of delivering meaningful, reliable, and scalable learning. This distinction is important because many learners remain excluded not only from schooling itself, but from quality educational experiences that produce strong and consistent outcomes (UNESCO, 2020; World Bank, 2025).

To evaluate current modes of education, the review applies six analytical criteria drawn from recurring themes in the literature: personalisation, decoupled delivery, structured guidance, accessibility, adaptability, and engagement. Together, these criteria provide a framework for assessing not only whether education is available, but whether it can be delivered in a way that is responsive to learner needs, less dependent on scarce professional expertise, and effective across varied educational contexts.

The review first sets out the criteria, then introduces the main modes of educational delivery and evaluates them against those criteria. It then identifies the gap that this evaluation reveals, before reviewing the evidence for structured and facilitator-supported learning that supports a facilitator-based model as a response to that gap. The same six criteria are used again in Chapter 6 to evaluate the platform built in this project.

3.2 Evaluation Criteria

Personalisation is an educational model's ability to respond to differences in learner ability, pace, and need. Personalised learning has been shown to improve engagement, achievement, and motivation by aligning instruction with individual learners (Hooshyar et al., 2024).

Decoupled delivery refers to the extent to which educational expertise can be separated from direct classroom instruction. This criterion captures how strongly the delivery of education depends on trained professionals. It is especially relevant in contexts facing teacher shortages, where reliance on qualified educators can constrain the reach, scalability, and resilience of education systems (UNESCO, 2024; Veine et al., 2022).

Structured guidance is the provision of clear instructional sequences, step-by-step lesson flows, and defined support materials that reduce ambiguity in delivery. Structured pedagogy has been identified as one of the most effective approaches for improving learning outcomes, particularly in low-resource settings where consistency of delivery is difficult to maintain (Piper et al., 2018; Snilstveit et al., 2016; World Bank, 2018).

Accessibility is an educational model's ability to overcome geographic, economic, and institutional barriers and make quality education more widely available. This remains a central concern in global education, where inequity is closely tied to poverty, remoteness, and limited resources (UNESCO, 2020; World Bank, 2025).

Adaptability is the capacity of an educational model to be adjusted across different learners, subjects, curricula, and contexts. This is important because learning environments vary significantly across age groups, institutions, and cultural settings, and rigid models often struggle to transfer effectively across them (Achuthan, 2025; Huang et al., 2026).

Engagement is the degree to which an educational model can sustain learner attention, participation, and involvement in the learning process. High levels of engagement are associated with stronger learning outcomes, while low engagement remains a common limitation of self-paced and online learning systems (Garrison et al., 1999; Jordan, 2015; Means et al., 2010).

3.3 Current Modes of Education

Six modes of education are considered. Together they cover how learning is organised today, from institutional schooling to informal and community-based provision.

Formal schooling is the dominant and most institutionally recognised mode of education. It is typically organised around structured curricula, trained teachers, age-based progression, and regulated assessment.

Homeschooling and self-directed learning take place outside an institution, led by families or learners themselves, who adapt content and pacing to their own goals and contexts.

Online and self-paced learning is delivered through platforms such as Khan Academy, Coursera, and Udemy, where learners work through digital content independently. A MOOC (Massive Open Online Course) is a free, web-based distance learning programme designed for a large number of geographically dispersed participants.

Synchronous and blended learning combine live instruction with digital content. A synchronous learning model is an educational approach where instructors and students interact in real time, engaging simultaneously from the same or different locations, through live video conferencing, instant messaging, or in-person sessions. Blended learning extends this by combining digital content with in-person or facilitated support.

Learning management systems and educational platforms such as Moodle, Canvas, Blackboard, and Google Classroom provide infrastructure for organising educational content, assessment, and communication.

Alternative and community-based models respond to settings where formal schooling is absent, weak or out of reach. Their shared feature is that instruction is delivered by community members who are not fully qualified teachers, as in para-teacher, volunteer-led and non-formal programmes. Pratham's Teaching at the Right Level groups children by their current learning

level and not by grade, and has been shown to raise learning when implemented well (Banerjee et al., 2017). BRAC's community-based schools are taught by women recruited from the local community, many with little formal education of their own, who are supported by monthly refresher training (Islam & Chowdhury, 2019).

3.4 Evaluation of Existing Models

Each model was assessed against the six criteria in Section 3.2, using the evidence discussed below. Table 2 summarises the result.

Table 2. Existing models of education compared against the six criteria

Model	Personalisation	Decoupled delivery	Structured guidance	Accessibility	Adaptability	Engagement
Formal schooling	Partial	Weak	Strong	Partial	Partial	Strong
Homeschooling and self-directed	Strong	Weak	Weak	Partial	Strong	Partial
Online and self-paced (MOOCs)	Partial	Partial	Weak	Strong	Partial	Weak
Synchronous and blended	Partial	Weak	Strong	Partial	Partial	Strong
Learning management systems	Partial	Weak	Partial	Strong	Strong	Partial
Community-based models	Partial	Strong	Partial	Partial	Partial	Partial

Ratings are qualitative judgements drawn from the literature reviewed in this section: Strong means the model meets the criterion well, Partial means it meets it in some conditions or to a limited degree, and Weak means it generally does not.

Formal schooling performs strongly on structured guidance, as it usually provides sequenced instruction, classroom routines, and clear expectations for learning progression (UNESCO, 2015). It also supports meaningful engagement through direct teacher-student interaction, peer discussion, and social learning environments. Personalisation is possible when teachers differentiate instruction, but this varies with class size, teacher workload, and available resources. The model performs less strongly on accessibility and decoupled delivery. It depends on infrastructure, staffing, and institutional coordination, which makes it difficult to establish and sustain in underserved regions, and its effectiveness is tied to the availability of qualified teachers, so expertise is not separated from delivery (UNESCO, 2024). Informal feedback from an academic coordinator at a school in Jaipur reflects the same dependence: "I can find five, six teachers only who really work for the children." Regulatory and administrative rigidity also slows its response to local needs and curricula. Formal schooling is strong in structure and legitimacy, but difficult to scale equitably across diverse settings.

Homeschooling and self-directed learning offer greater flexibility and, in principle, strong personalisation and some adaptability, as content and pacing can follow the learner's needs and interests. However, their effectiveness depends heavily on the learner's capacity for self-regulation and on the quality of support at home. Research on self-directed learning in digital settings suggests that learners need strong planning, monitoring, and self-management skills to benefit from independent learning systems (Chen, 2009; Kim et al., 2014). For younger learners this often results in weak structured guidance and inconsistent engagement unless a capable adult is present. Accessibility is uneven, since success depends on parental time, educational confidence, and access to quality resources, so these models may widen inequities rather than reduce them. They also do little to address decoupled delivery, because provision still depends

on the capability of the adult directly supporting the learner. The literature does not support them as a broadly scalable response to the global shortage of quality education.

Online and self-paced learning has greatly expanded access to educational content. It performs strongly on accessibility, reducing geographic barriers and making materials available at low cost and large scale, and it supports some personalisation where pacing is flexible or content is recommended. However, the literature consistently shows that self-paced online learning struggles with engagement and structured guidance. Jordan (2015) found that MOOC completion rates were often below 15%, suggesting that learners frequently disengage when left to navigate learning independently, and Kizilcec et al. (2013) similarly found that sustained participation is limited. Such systems also shift the burden of learning onto the student rather than reducing dependence on teachers through structured facilitation, which makes them especially problematic for younger learners who typically need stronger instructional support and supervision.

Synchronous and blended learning seek to preserve the interactive qualities of classroom teaching. Synchronous models generally engage learners more than self-paced learning, because learners can interact with instructors and peers in real time, and the Community of Inquiry framework emphasises the role of teaching presence, social presence, and interaction in meaningful online learning (Garrison et al., 1999). Blended learning combines flexibility with human interaction and can support more effective structured guidance when digital content is embedded within planned teaching sequences; Means et al. (2010) found that blended environments often outperform purely online or purely face-to-face formats. However, both remain limited in decoupled delivery, since they still depend on skilled teachers or facilitators being present in real time or in person. Their accessibility is also affected by infrastructure requirements such as reliable internet, scheduling coordination, and device access, and their implementation often requires substantial planning and institutional support.

Learning management systems support accessibility by making resources available digitally, and adaptability, since content can be rearranged, reused, or modified across subjects and classes. Despite this, they are relatively weak on structured guidance when considered as stand-alone educational models. Coates et al. (2005) found that LMSs often function more as administrative repositories than as active pedagogical systems, with their effectiveness depending largely on how teachers design lessons and manage delivery around them. They support the organisation of education rather than the act of teaching, and they assume the presence of teacher expertise to interpret content, lead activities, and sustain engagement. LMSs are therefore useful tools, but they do not by themselves resolve the central challenges of quality, consistency, and teacher dependency.

Alternative and community-based models begin to challenge the assumption that all effective education must be delivered by fully qualified teachers. They often perform better on decoupled delivery, as they demonstrate that educational provision can, under certain conditions, be delivered by community members or minimally trained facilitators. Initiatives such as Pratham's Teaching at the Right Level (Banerjee et al., 2017) and BRAC's community-based schools, which rely on locally recruited teachers with regular training (Islam & Chowdhury, 2019), suggest that non-traditional delivery can be effective when supported by strong programme structure and targeted pedagogical design. Where content and routines are centrally defined, they also perform relatively well on structured guidance, and their local nature may support accessibility in communities where formal schooling is weak or absent. However, adaptability and engagement vary with how each programme is designed and implemented, and many community-based models are programme-specific rather than delivered through reusable, scalable digital systems. This limits their transferability, although they provide important evidence that educational expertise and delivery can be partly separated when strong support structures are in place.

Across the literature, a consistent pattern emerges: existing educational models perform strongly on some of the criteria but poorly on others. The central limitation is not that current models are ineffective in all respects, but that they do not balance the full set of needs implied by the global education crisis. Formal schooling offers strong structured guidance and often strong engagement, but is highly dependent on qualified teachers and difficult to extend into underserved settings. Self-paced online learning performs strongly on accessibility and in some cases personalisation, but poorly on engagement and structured guidance. LMSs support adaptability and resource organisation, yet do not provide guided delivery or reduce dependence on educators. Alternative community-based models begin to address decoupled delivery, but are often programme-specific and not implemented through scalable platform systems. Current systems also rarely combine decoupled delivery with strong structured guidance. Where expertise is separated from delivery, this often happens through self-paced content consumption rather than supported classroom facilitation, and where structure and engagement are strong, this usually depends on a qualified educator being present.

3.5 Identified Market and Research Gap

The literature suggests that current educational models tend to divide into two broad groups: those that provide quality through expert-led, resource-intensive delivery, and those that provide access through low-cost, content-based delivery. What remains underdeveloped is a model that combines accessibility, structured guidance, engagement, and decoupled delivery in a coherent and scalable form. In particular, there is limited evidence of mainstream educational platforms designed specifically to guide non-expert facilitators through live lesson delivery using expert-created instructional sequences.

This gap is significant because it points to the possibility of a different educational architecture. If lessons can be centrally designed with strong pedagogical structure, then classroom delivery may not need to rely entirely on expert teachers. A facilitator-based model may therefore offer a way to preserve human interaction and classroom engagement while improving accessibility and reducing professional bottlenecks. From this perspective, the value of the proposed model lies not simply in digitising education, but in reconfiguring the relationship between expertise, delivery, and access. Such a model must let experts design lessons in advance, guide a facilitator through delivering them, work across subjects and contexts, and keep the human presence that supports engagement. These requirements shape the objectives set out in Chapter 2 and the research approach in Chapter 4.

3.6 Evidence for Structured and Facilitator-Supported Learning

Whether a facilitator-based model could close this gap depends on whether well-structured lessons can be delivered effectively by people who are not fully qualified teachers. The literature on structured pedagogy provides strong support for the value of clear and well-sequenced instructional design. Piper et al. (2018) found that structured pedagogical programmes in Kenya produced substantial gains in literacy, while Snilstveit et al. (2016) identified structured pedagogy as one of the most effective categories of intervention in low- and middle-income countries. The World Bank (2018) similarly recommended structured lesson plans and teacher support as high-impact strategies for improving foundational learning. These findings suggest that structured guidance is not a minor design feature, but a core contributor to educational effectiveness.

This is particularly relevant when considering decoupled delivery. If lesson content, teaching steps, and support materials are clearly designed in advance, it becomes more plausible that classroom delivery can be supported by people who are not fully qualified teachers. Research on facilitators and learning assistants supports this possibility. Veine et al. (2022) found that learning assistants can contribute meaningfully as facilitators when they are supported with training and structured responsibilities. Daga-As and Guntalidad (2025) similarly showed that

alternative learning system facilitators can produce meaningful educational impact despite limited formal training, provided appropriate support systems are in place. The community-based programmes discussed in Section 3.4 apply the same principle without a digital platform.

Facilitator-supported learning may also improve engagement, particularly in comparison with self-paced digital learning. Human presence in the classroom can support discipline, encouragement, and participation, even where the facilitator is not the original designer of the lesson. In addition, facilitator-based models may improve accessibility by widening the pool of individuals capable of delivering education, especially in communities where fully qualified teachers are scarce. This does not eliminate the need for expertise; rather, it changes where that expertise is located. Instead of being concentrated entirely in the classroom teacher, expertise can be embedded in lesson design, instructional sequences, and central content creation.

The same studies also attach conditions. Facilitators succeeded where they were trained, given defined responsibilities, and backed by support systems, and a structured programme benefits learners only when it is delivered as designed. This body of literature does not prove that all facilitator-based models are equally effective, nor does it suggest that expertise becomes unnecessary. However, it does provide a theoretical and empirical basis for exploring models in which educational design and educational delivery are more clearly separated, and it identifies the condition such a model must meet: the facilitator must be supported well enough to deliver the lesson as the expert intended.

3.7 Conclusion

This review has shown that the global shortage of quality education is a substantial and multidimensional problem, reflected in widespread learning poverty, persistent teacher shortages, and major inequalities in educational access and quality. Existing modes of education respond to this challenge in different ways, but none fully satisfies the combined demands of personalisation, decoupled delivery, structured guidance, accessibility, adaptability, and engagement.

The literature therefore suggests a clear gap for a model that combines the strengths of structured pedagogical design with a more flexible and accessible delivery model. A facilitator-based approach, in which lessons are designed by experts and delivered through structured support by non-expert facilitators, appears well positioned to address this gap. As such, it provides a plausible response to the limitations identified across existing educational models and offers a strong conceptual basis for the present project. The six criteria developed here are carried forward as the standard against which the platform is evaluated in Chapter 6.

4. Research Approach

4.1 Research Question and Sub-Questions

As mentioned previously this report addresses the research question: *How can a software platform be designed so that any approved facilitator, regardless of subject expertise, delivers an expert-designed lesson as it was intended?* The question is broken into three sub-questions (Table 3).

Table 3. Sub-questions, how each is answered, and where it is reported

Sub-question	How it is answered	Reported in
SQ1 Authoring. What does an admin need from a platform to capture a lesson's content, sequence and delivery instructions, so that it can be run without them in the room?	Design of the block-based authoring model; a check that it can represent lessons from real curricula; testing of the worksheet generator on sample lessons; the time taken to build a lesson	Chapters 5 and 6
SQ2 Delivery. How can the platform control who delivers a lesson and guide them through it, so that delivery follows the admin's design?	Design of approval, roles and the walkthrough; technical tests of the role and lesson-flow rules; a heuristic review of the interface; informal feedback	Chapters 5 and 6
SQ3 Fit. How well does the platform meet the needs identified in the literature and by practitioners, and how does it differ from existing systems?	Assessment against the six criteria from Chapter 3; a benchmark against existing systems; informal feedback from practitioners	Chapters 6 and 8

The first two sub-questions concern what was built, and are answered through design reasoning and testing. The third concerns how the result compares with what already exists, and is answered against the same six criteria used to review the literature.

The project's main output is a working proof-of-concept platform, and its contribution is the research that was conducted, an account of how that platform was designed and the platform's evaluation against various frameworks and criteria. Claims are limited to what was built and tested. No claim is made about learning outcomes, because classroom trials are outside the scope set in Section 2.5.

4.2 Overall Approach

The project followed five stages: understand the problem, design a solution, build it, evaluate it, and refine it (Figure 2). The stages were not run once in sequence. The build was carried out in short cycles, and feedback gathered at the end of each cycle was used to revise the design before the next one began. This suits a project whose aim is a working platform: the design could be tested against real use early, instead of being fixed in advance.

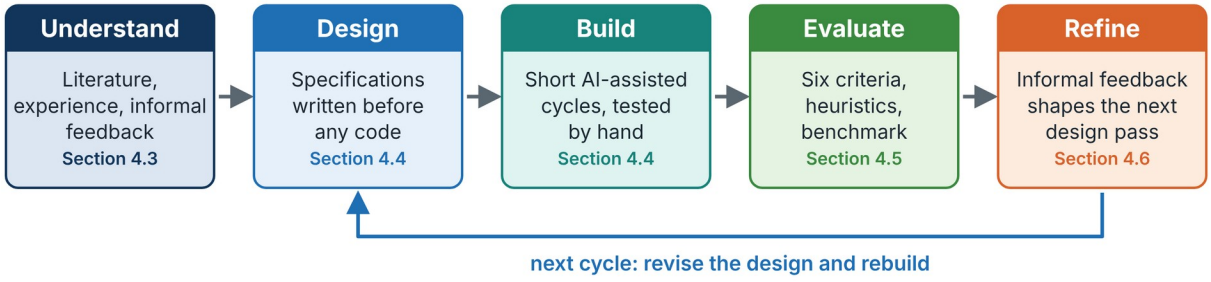


Figure 2. The research approach. Each stage is described in the sections that follow.

4.3 Understanding the Problem

The problem was approached from four directions, so that no single source carried the whole argument.

The first was the literature. The review in Chapter 3 examined how existing modes of education perform, derived the six criteria used to compare them, and gathered the evidence for structured and facilitator-supported learning. It shaped the problem statement in Section 2.2 and the objectives in Section 2.4.

The second was my own experience as a student in Jaipur, described in Section 2.1. This provided the starting observation that the quality of learning depends heavily on the individual teacher.

The third was informal chats with practitioners. During the project, a conversation with my high school's academic coordinator tested whether the problem seen in the literature matched what school staff experience day to day. It surfaced the points that now shape the design: teaching quality depends on the individual teacher, a departing teacher disrupts the whole school, and no teacher should be sent into a classroom unprepared. The conversation was used to inform design decisions and not as research data to be generalised from.

The fourth was the use of AI tools to find relevant papers. AI was used as a search aid, to suggest sources and search terms. Every paper found this way was then opened and checked by hand before use: that the source exists, that it supports the claim attributed to it, and that its details are cited correctly. How AI was used more widely in the project is described in Chapter 7.

4.4 Designing and Building

The platform was designed and built specification first. Before any code was written, a project document set out the problem definition, the goals, how the system should work, what each page needs and how it should look, along with what would be built and with which tools. A coding-feasibility pass checked that it could be built as described. Each feature then got its own specification document, written before that feature was built.

Each feature specification broke the work into small pieces that can be thought of as short AI-assisted coding sprints, in the manner of agile development (Figure 3). Each session was built with an AI coding assistant and then tested by hand in real time before the next one began, so that a fault was found while its cause was still recent. The specification was not treated as fixed: it continued to be refined as the code was written and as testing showed what had been missed.

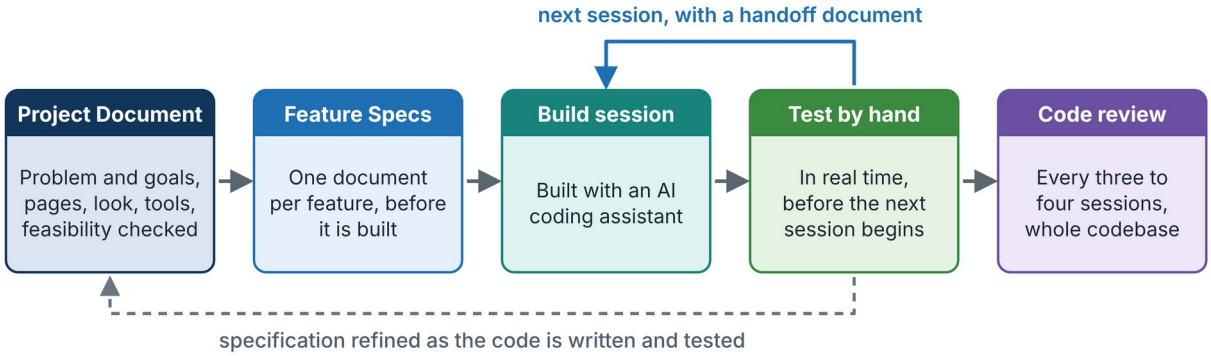


Figure 3. How the platform was designed and built: specification first, then short build sessions, each tested before the next.

4.5 Evaluating Against Existing Frameworks

When the platform was built, it was evaluated through various frameworks and criteria. This keeps the evaluation open to challenge and ties it to the literature.

The six criteria from Chapter 3 were used to assess the platform and to benchmark it against existing systems. The platform is rated on the same Strong, Partial and Weak scale used in Table 2, so that it can sit directly beside the existing modes, and the benchmark compares it with systems chosen to represent the modes reviewed in Chapter 3: learning management systems (Moodle, Google Classroom), self-paced platforms (Khan Academy, Kolibri) and structured, facilitator-supported provision (LEAD School, Bridge and Pratham).

The interface and design were tested through a heuristic review against Nielsen's usability heuristics (Nielsen, 1994), with each problem found given a severity rating, and through the informal feedback described in Section 4.6.

Technical tests checked that the rules the design depends on hold: for example, that a facilitator cannot reach an admin action, that a lesson cannot be opened out of order before its first

walkthrough, and that pages respond within acceptable times. Authoring tests checked whether what the platform produces resembles what a lesson needs: coverage of real curriculum lessons, the worksheet generator on sample lessons, and the time taken to build a lesson.

These methods have limits. The ratings are qualitative judgements and not measurements, the benchmark rests on published descriptions of the other systems, and none of the methods measures learning outcomes. Claims in Chapters 6 and 8 are kept to what these methods can show.

4.6 Informal Feedback to Improve the Design

During the project, informal feedback was gathered from a range of people whose perspectives bear on the platform: practitioners with school and classroom experience, a student and tutor who reviewed the interface, and peers. The aim was to improve the design, not to produce findings that generalise.

The feedback was collected through informal conversations and demonstrations of the platform, notes from the chats were kept and later linked to the design decision it supported, changed or challenged (Chapter 6). This keeps the link between what people said and what changed in the platform traceable.

Because the feedback is informal and comes from a small number of people, it is used to support design decisions and to check the fit of the platform. It is not treated as evidence that the platform improves learning. Its limits are discussed in Chapter 8.

5. System Design

5.1 Design Principles and Structure

The design follows from the research question. If an approved facilitator is to deliver a lesson as it was intended, the expertise has to be built into the lesson and the platform, and not supplied by the person delivering it. This gives the platform two roles. Admins, who have teaching and subject expertise, create and organise lessons. Facilitators, who are physically present in class with the students, deliver those lessons through a guided, step-by-step interface but cannot edit them. The split follows from the problem being solved: facilitators are assumed to have no subject expertise, so authoring stays with admins, who bring the judgement to design a lesson well, and facilitators carry out that design instead of making instructional calls on the fly. This follows the evidence in Section 3.6 for structured, facilitator-supported lessons.

Using the platform takes five steps (Figure 4). An admin builds a lesson from blocks. A facilitator requests access, choosing the classes they want to teach. An admin reviews the candidate, approves the request, and gives access to the requested classes in the same step. The facilitator then walks through the lesson once, with each block unlocking only after the one before it. This is done because informal feedback from my high school's academic coordinator (Section 4.3) suggested that teachers need time to familiarise themselves with content before teaching it, so the walkthrough gives the facilitator that time and a chance to prepare any materials the lesson needs. Questions are expected during the walkthrough, which is why the platform also has a chat that connects the facilitator directly with the admins. Finally, the facilitator runs the lesson with the class and can move through it freely. The order is deliberate. A lesson is completely authored from start to end before anyone delivers it, so everything a facilitator needs is already in it. It is also important to consider that not anyone can be a facilitator, that is why access to the platform is requested and then approved, so that an admin can control who delivers lessons.

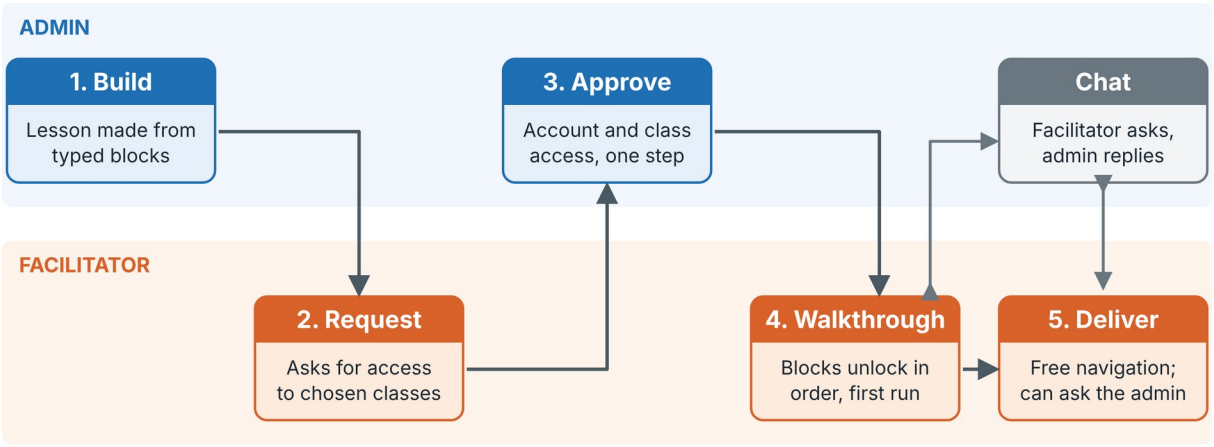


Figure 4. The five steps of using the platform. Admins build and approve, and facilitators request, walk through and deliver. The chat connects facilitators to admins during the walkthrough and delivery.

Content is neatly organised within the platform and follows a fixed hierarchy: Class, Subject, Lesson and Block. It mirrors how a school already organises teaching, by year level, then subject, then individual lessons, which matches the mental model of someone responsible for a consistent group of students working toward a shared understanding, and not an abstract content taxonomy. A lesson is an ordered sequence of blocks (Section 5.3), so the structure of the lesson itself carries guidance a facilitator would otherwise need subject expertise to supply.

5.2 Technology Stack, Architecture and Data Model

The platform is a web application built with Next.js 14 and TypeScript, so a facilitator needs nothing more than a browser. Next.js pairs Server Components, which read data, invokes Server Actions that make every change. This removes the need for a separate API layer and keeps each permission check next to the logic it protects. Middleware checks the session and route before a request reaches the application. Supabase supplies login, the Postgres database and private file storage from a single vendor, giving one integration to maintain instead of three, and Prisma sits between the application and the database. Resend sends the invitation emails, and the app is deployed on Vercel, where every push to the main branch is built and deployed automatically.

In this tech stack we made three choices that involved a trade-off. First, Prisma was pinned below version 7 and not migrated, because version 7 removed syntax the schema depended on. This was done because version 7 removed syntax the schema depended on, there was a lot of available documentation below version 7 and also because stability was prioritised ahead of being current. Second, Resend was used for real invitation emails and not a mock, so that the approval flow could be tested end to end, and the invite link is also shown on the admin page so approval still works when email is not set up. Third, Vercel's functions were pinned to Sydney, next to the database. Selecting Sydney instead of the default USA as our region for the Supabase and Vercel servers cut the time taken to add a block from about 10.7 seconds to about 434 milliseconds.

Figure 5 shows how these parts fit together. A request from either role passes through middleware to Server Components, which read from Supabase, or Server Actions, which write to it. Server Actions also call Resend when an invitation is sent and the Claude API when a worksheet is drafted (Section 5.6).

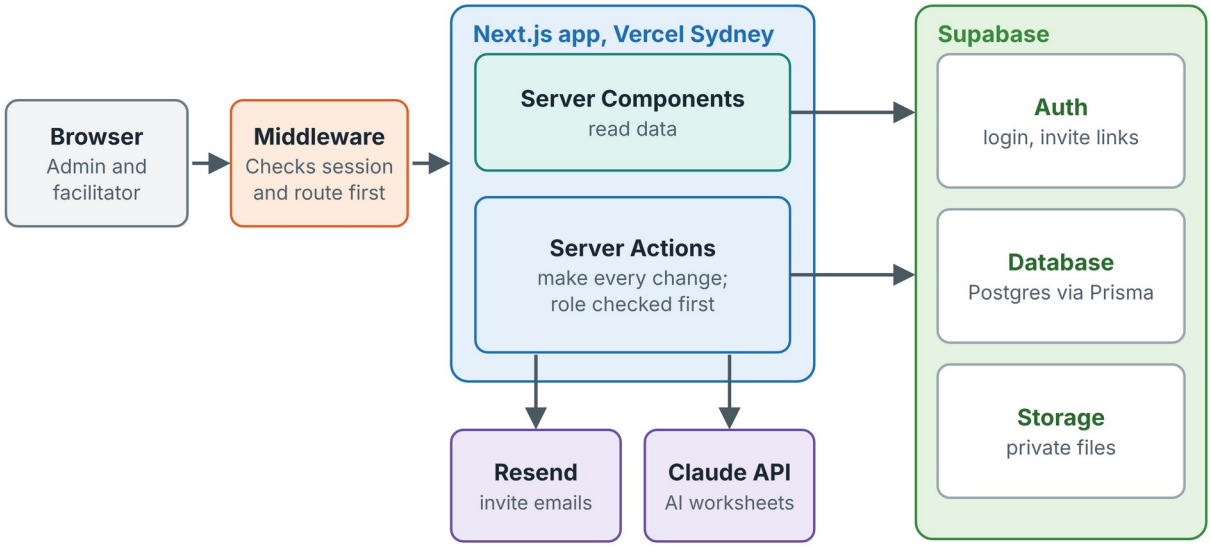


Figure 5. System architecture. Requests pass through middleware to Server Components and Server Actions, which use Supabase; Server Actions also call Resend and the Claude API.

The data model has eleven models (Figure 6). Content follows the hierarchy from Organisation down to Block. A lesson is either a draft or published and has a duration. A block has a type, an order and its content, and that content is held in one flexible JSON field rather than in a separate table for each of the nine block types. This keeps the model simple and easy to extend, at the cost of weaker compile-time checking of block content, a trade-off that was accepted and documented in the code. Around the content sit five other kinds of records. Users are either admins or facilitators. Assignments give a facilitator access to a class and subject. Progress records are kept one per facilitator per lesson, are what the delivery flow depends on (Section 5.4). Sign-up requests hold a request until it is approved, so no account exists before then. Messages and notifications carry communication between admins and facilitators. There is no student record: students do not have accounts, so the platform holds no student data.

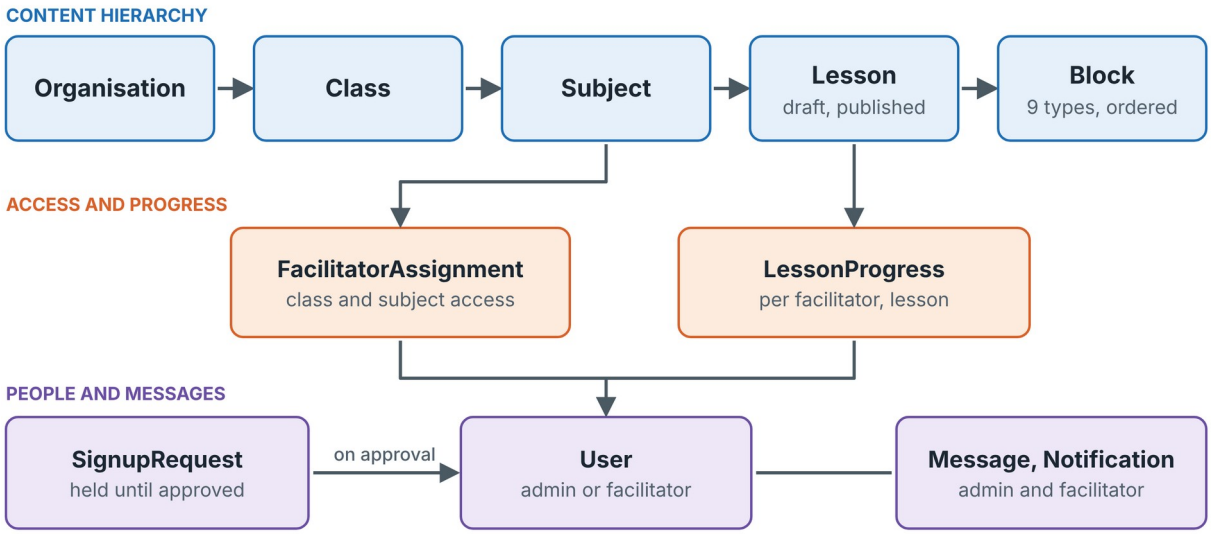


Figure 6. The data model: eleven models, with access and progress records tied to users.

5.3 Lesson Authoring

The platform is built around the concept of blocks. A lesson is an ordered sequence of blocks, each one a single step, and there are nine types: video, workbook, activity, worksheet, reading, quiz, discussion, break and note. A single upload box or one long document would have been simpler to build, but it would carry no guidance, and the facilitator would have to decide for

themselves what happens next. Typed blocks avoid this because each type has fields suited to it (for example, a URL and duration for a video, or an uploaded file for a worksheet), and their order tells the facilitator what to do and when. This is what the step-by-step delivery in Section 5.4 relies on. Admins assemble lessons with a drag-and-drop block builder (Figure 7).

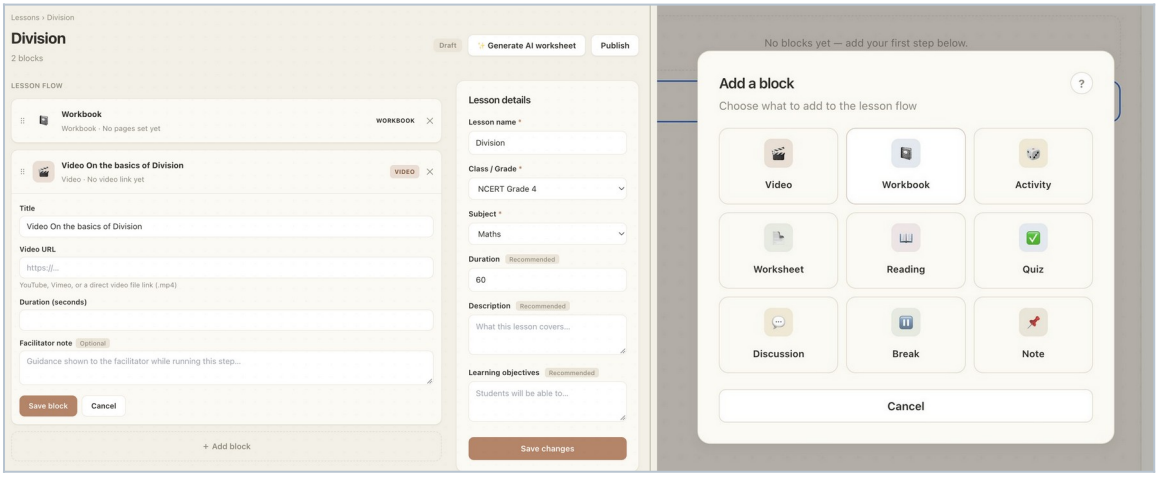


Figure 7. The lesson authoring page (a) and the block picker (b). Blocks are added from the picker, reordered by dragging and edited in place, and the lesson stays a draft until it is published.

Every lesson starts as a draft, so an admin can test it before a facilitator sees it, and publishing is not final, because a lesson can be edited or hidden afterwards which allows the system to be flexible and responsive to any feedback. The platform was seeded with about 110 lessons, based on the NCERT curriculum for Grades 1 to 4 in India and the New Zealand Curriculum for Years 1 to 4, in maths and science and covering one or two core concepts. We also implemented a curriculum filter to make navigation easy and ensure that the tools could be tried on realistic content from two countries.

Feedback on the admin panel changed how this structure is navigated. Classes had been a flat list, and Lessons a single table repeating the same information, so there was no way to get from a class to its lessons. Classes became a drill-down from class cards (Figure 8) to subject cards to the lessons in that subject, with a breadcrumb at each level and a Create lesson shortcut that pre-fills the class and subject. Lessons became a separate, searchable index grouped by class and subject, with a status filter. The two pages now do different jobs, and the drill-down follows the hierarchy in Section 5.1, which fits the heuristic of matching the system to the real world, while the pre-filled shortcut supports flexibility and efficiency of use (Nielsen, 1994).

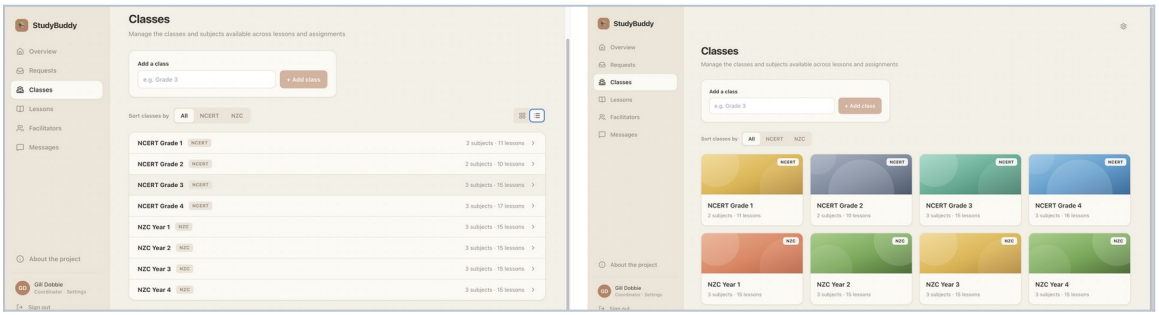


Figure 8. The Classes page in (a) list view and (b) card view, which an admin can switch between. Classes can be filtered by curriculum, and each one opens into its subjects and lessons.

The same structure held up when the platform was reviewed with a medical student and tutor who had used several online learning platforms. In a guided session, she found a lesson by going from class to subject to lesson without help, and said the curriculum filter made lessons easier to browse. Her one change to authoring was that the button for removing a block should say Delete and not Confirm, which follows convention and makes the action clearer (consistency and standards, and error prevention). Removing a block takes two clicks and no pop-up, which

guards against accidental deletion without interrupting every removal, and the button now reads Delete.

5.4 Delivering a Lesson

When a facilitator opens a lesson for the first time, only its first block is available. Finishing a block brings up a short prompt asking whether to move on or to ask the admin a question, and the next block unlocks only once the one before it is finished. When every block has been finished, the lesson stays fully unlocked, so the facilitator can move through it freely while teaching. A step list shows which blocks are done, which is current and which are locked (Figure 9), so the facilitator never has to guess what comes next (visibility of system status; Nielsen, 1994).

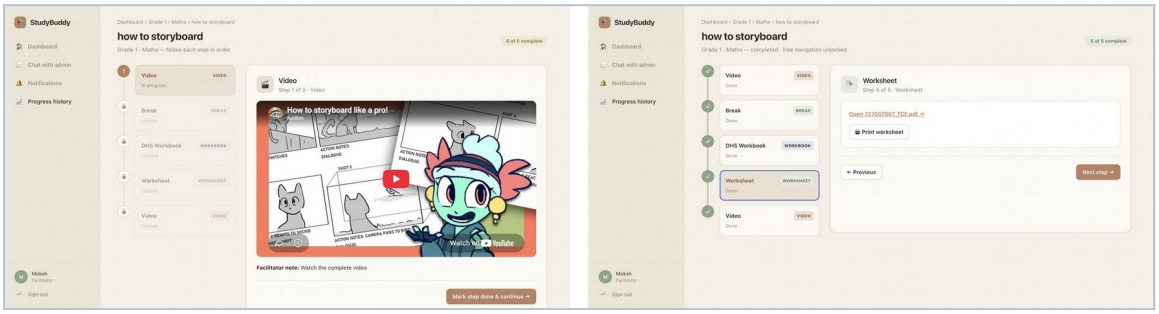


Figure 9. The lesson player (a) when first opened, with only the first block available and the rest locked, and (b) after every block is finished, when the lesson stays fully unlocked.

The state of each block is never stored or trusted in the browser. It is worked out on the server each time from one progress record per facilitator per lesson, and finishing a step is checked there again. Tracking progress in the browser would have been quicker to build, but the order could then be skipped by changing what the browser sends, and progress would be lost when the page is refreshed or the facilitator signs out. With the record held on the server, a refresh or a new sign-in resumes at the same block, making the system more reliable and fail proof.

In the conversation with the medical student and tutor, the locked steps were understood straight away without explanation, and she thought the lesson player needed no changes. Her feedback on the pages around it led to three changes: lesson cards now say what their step count and time meant, a back button returns from a lesson to the lesson list, and the chat shows an expected reply time or when a message was seen. These follow the heuristics of matching the system to the real world, user control and freedom, and visibility of system status (Nielsen, 1994).

5.5 Security and Access

The platform puts lessons in front of students through people without subject expertise, so who gets access and what they can do is controlled deliberately and not left to convenience. Accounts follow a request-first model. A person who signs up creates a request that waits for an admin, and no account exists until the admin approves it. The request names the classes the person wants to teach, so access to those classes is granted in the same step, and the approved facilitator receives an invitation link. Open self-registration, the alternative, would give no control over who ends up delivering lessons. The medical student and tutor both read this step as a controlled system where nobody can simply sign in, which is the intent.

Every admin action is checked on the server and not only hidden in the interface. This was added after a real gap was found during the build: protection existed only in the interface, so a facilitator who guessed an admin address could have reached admin actions. Changing a password requires the current one, which blocks a silent change. The tutor could not find how to remove a facilitator’s access, because “Assign” suggests adding only, so the action is now “Manage access” (recognition rather than recall; Nielsen, 1994).

The platform holds no student data. Students do not have accounts, and the only people recorded are admins and facilitators, with their class assignments and lesson progress. Uploaded files such as worksheets are stored privately and reached through signed links.

5.6 The Worksheet Generator

Once a lesson has been created, an admin will often want supplementary learning material to go with it, which is why the platform includes an AI worksheet generator. It is the one place the platform uses AI to produce content, so the design question was how far to trust it. The answer was that the AI drafts and an admin approves, twice. The generator first reads everything in the lesson, including documents, videos and activity instructions, and writes a summary. The admin checks and can edit this summary, then chooses the type of worksheet they want on the generator page. The worksheet and its answer key are then written from the approved summary in an editable form, and the admin can edit, regenerate or reformat them. When the admin is satisfied they click confirm, and the worksheet is added to the lesson as a new block (Figure 10).

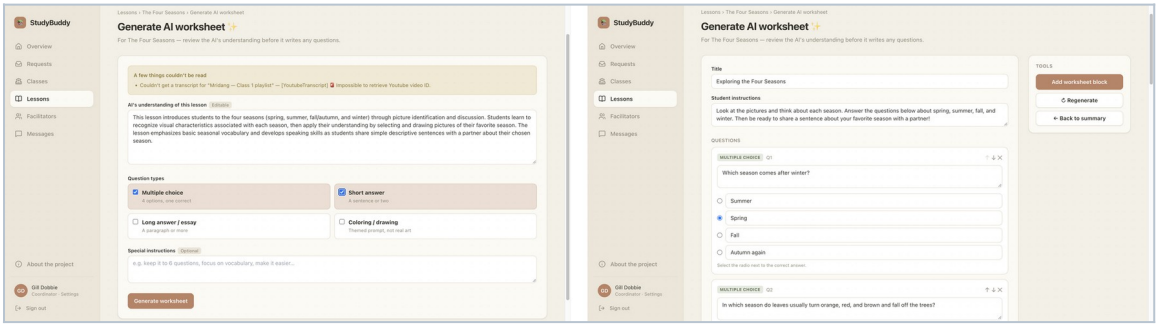


Figure 10. The worksheet generator. In step (a) the admin reviews the AI's summary of the lesson, which is editable and lists anything that could not be read, and chooses the question types. In step (b) the generated worksheet is edited, regenerated or added to the lesson as a new block.

A smaller model (Claude Haiku 4.5) does the summarising and a larger one (Claude Sonnet 5) writes the worksheet. The worksheet is requested through forced tool use, so it comes back in a fixed structure that can be saved as a block and not as free text that has to be parsed. The generator considers the lesson as a whole and summarises it based on the blocks like YouTube captions and the text of uploaded PDF and DOCX files. If a source cannot be read, the admin is told in plain words before anything is generated, so a worksheet is never quietly built from part of a source. The answer key is always produced and can be edited by the admins. This ensures reliability and the verified answer key helps the facilitator the most. Regenerating works from the saved summary, so the sources are not read again, and a generated worksheet never overwrites an existing block. When no API key is configured the feature shows a "not configured" state and the rest of the platform is unaffected.

5.7 Interface Design and Feedback

The admin interface was redesigned as an iteration of my own, guided by my judgement and by Saroj's review. Instead of committing to one idea, I built six full mock-ups on the same real class and lesson data, in different styles (a developer console, an enterprise style, a soft productivity style, Material Design 3, a notebook style built around school subjects, and a refinement of the existing palette), so the choice could be made by looking at them. Three were kept and are now themes the admin can switch between. Each theme is a set of design tokens for colour, radius, shadow and type that drives the same shared components, so changing theme does not mean rebuilding any page, which supports consistency and standards (Nielsen, 1994). The platform now offers two navigation layouts, a top bar and a side bar, to better suit the user's preference and ease of access. Emojis in the previous design were replaced with icons that look the same on every system and recolour with the theme. Class cards were inspired by the Canvas course cards, with a banner image, because Canvas is the learning platform people are most likely to

know already (match between the system and the real world). The theme and layout settings were grouped and put under the settings icon at the top right corner of the screen contributing to an aesthetic and intuitive design.

6. Evaluation

6.1 Evaluation Approach

The evaluation asks whether the platform does what the design claims, not whether students learn more from it. Answering that would need a classroom study with real facilitators and students, which was outside the scope of this project, and it is listed as future work in Chapter 9. Five things were tested instead. The platform was benchmarked against existing platforms on the six criteria from the literature review (Section 3.2) and reviewed against Nielsen's ten heuristics (Nielsen, 1994). Informal design feedback was gathered from three people, technical tests covered access, lesson flow and speed, and authoring effort and the worksheet generator were manually assessed. Together they answer the three sub-questions of the project. The benchmark and the design feedback show how well the platform fits the problem it set out to solve. The heuristic review, the feedback and the technical tests show how usable and reliable the system is. The authoring and worksheet assessments show how practical it is to build lessons using the platform at scale. The limits of this approach are stated in Section 6.7.

6.2 Benchmark Against Existing Platforms

The six criteria from the literature review (Section 3.2) were used to evaluate StudyBuddy and, in the same way, to benchmark it against existing platforms (Table 4). Platforms that are built around the same core concept tend to behave alike, so they are grouped by that concept and not listed one by one: learning management systems (Moodle and Google Classroom), self-paced content platforms (Khan Academy and Kolibri), and programme-based models (LEAD School, Bridge International Academies and Pratham). The grouping follows the modes of education rated in Table 2, with StudyBuddy added. Each rating is Strong, Partial or Limited, and is based on public descriptions of each platform and hands-on experience with them.

Table 4. Existing platforms and StudyBuddy rated against the six criteria.

Kind of platform	Personalisation	Decoupled delivery	Structured guidance	Accessibility	Adaptability	Engagement
Learning management systems (Moodle, Google Classroom)	Limited	Limited	Limited	Partial	Strong	Partial
Self-paced content (Khan Academy, Kolibri)	Partial	Partial	Limited	Strong	Partial	Limited
Programme-based (LEAD School, Bridge, Pratham)	Limited	Strong	Strong	Partial	Limited	Partial
StudyBuddy	Limited	Strong*	Strong	Partial	Strong	Partial*

*Rated from the design, not from classroom use.

StudyBuddy is the only row that combines strong decoupled delivery, strong structured guidance and strong adaptability. Programme-based models reach strong decoupling and guidance, but they are tied to one organisation's curriculum and delivery, which limits their adaptability and personalisation across varying circumstances. Learning management systems

are adaptable and widely available, but both assume a teacher who designs and runs the lesson, so neither reduces dependence on expertise. Self-paced platforms give access to content at scale, but expect the learner to work through it alone, and none has a walkthrough that guides a person in the room.

The comparison also shows where StudyBuddy falls short. Accessibility is partial because the platform needs a device and an internet connection, whereas Kolibri is built to work offline. It has limited personalisation, since every student in a class sees the same lesson, and engagement depends on the quality of the lesson content and on the facilitator. These two gaps are why the second half of the project, the learner insights, was developed (Thazin, 2026). Similar to this part Its use in a real classroom has not yet been tested, but if it is used as intended it should make the platform more personalised and more engaging.

6.3 Heuristic Review

I reviewed the admin and facilitator sides against Nielsen's ten heuristics (Nielsen, 1994) and recorded how the platform responds to each (Table A1, Appendix A).

The review showed that each heuristic is met by at least one specific feature of the platform. A second, more critical pass, run with an AI assistant, then looked for problems the first had missed and rated each by severity on Nielsen's scale of 0 to 4. It found a few problems, mostly minor or cosmetic, and the most notable are in Table 5.

Table 5. Examples from the second heuristic review.

Heuristic	Problem found	Severity
1. Visibility of system status	The facilitator lesson list shows "Not started" for every lesson	3
4. Consistency and standards	Labels on the login and sign-up forms are not tied to their inputs	3
8. Aesthetic and minimalist design	Contrast in the default theme is below the WCAG AA level for muted text and the main button	3
1. Visibility of system status	Browser tab titles are the same on every page	2

These interface improvements were considered and resolved later while refining the system further.

6.4 Informal Design Feedback

Three people gave feedback that shaped the design. The academic coordinator at my school (Section 4.3) gave informal feedback on the problem and not on the platform, so her input mainly informs the design reasoning in Chapter 5. My friend Saroj, who has served as an admin in many organisations, reviewed the admin panel, which fed the redesign in Section 5.7. The most detailed review of the platform itself came from a my friend who wishes to stay anonymous, she is a medical student and tutor who has used several online learning platforms. The chat lasted about 20 minutes and I observed her navigate the platform independently

The reviewer found a lesson on her own by going from class to subject to lesson, understood the locked steps without explanation, and read the approval step as a controlled system that nobody can simply sign in to. She also described the layout as clear and uncluttered. She raised six problems: the labels on lesson cards, the way back to the lesson page from the lesson editor, how to remove access, the "Confirm" button, low contrast between cards and background, and how fast an admin replies. Each led to a change, described in Chapter 5 and not repeated here. The feedback was used as design input, and used to guide my design decisions later in the project.

6.5 Technical Tests

Technical tests covered access and security, the rules of the lesson flow, and the speed of common actions. The tests were run on the deployed platform and also on a local build to ensure consistent performance. A summary of these tests is provided below in Table 6.

Table 6. Access and security checks.

Check	Result
Open 20 protected routes without signing in	20 of 20 redirected to login (local build and live site)
Send 19 forged admin actions with a facilitator session	19 of 19 rejected, no data changed
Open lessons, classes or subjects not granted (8 cases)	8 of 8 not found
Open or progress a draft lesson in a granted subject	Hidden and rejected
Sign in before approval or before using the invite (7 attempts)	0 of 7 worked
Use an altered, reused or genuine invite link	Altered and reused rejected, genuine accepted once
Change password in Settings with a wrong or missing current password	Rejected
Request a stored file without a signed link	Rejected; signed links expire after 3,600 s
Student data stored	None; staff data only

The rules the design depends on held where they matter most (Table 6). Every protected route turned away a visitor who was not signed in, all 19 forged admin actions sent with a facilitator's session were rejected and changed nothing, and a facilitator could not open any lesson, class or subject they had not been granted. Draft lessons stayed hidden, a person with a pending request could not sign in, altered or reused invite links were rejected, and stored files were reachable only through links that expire after an hour. The system holds no student data.

The lesson-flow rules held in 13 of 14 checks (Table 7). Only the first block opens on a first visit, the content of locked blocks never reaches the browser, finishing a block unlocks exactly the next, progress survives a refresh and a sign-out, and every forged skip-ahead call was rejected. The one failure is that progress is stored as a position, so deleting a finished block while a facilitator is mid-lesson can make them skip a step.

Table 7. Lesson-flow rules on a five-block lesson.

Check	Result
Only block 1 unlocked on first open; locked content not sent to the browser	Pass (0 of 4 locked steps leaked)
Finishing a block unlocks exactly the next one	Pass
Progress kept after refresh, and after sign-out and sign-in	Pass
Completion page opened before the last block	Redirected back into the lesson
Last block done completes the lesson and unlocks everything	Pass
Forged skip-ahead and cross-lesson progress calls (4)	4 of 4 rejected
Admin deletes a finished block mid-lesson	Failed: facilitator skips a step

On speed, every page and action had a median response under 1.2 seconds in 25 runs each (Table A2, Appendix A), and a 25-step lesson loaded as fast as a 5-step one (1.14 s against 1.15 s), so lesson length does not slow the player. On the live site, public pages answered in a median

of 61 to 88 ms from Auckland. The other timings come from a laptop in Auckland with the database in Sydney, the reverse of production, so they are indicative and not production figures.

6.6 Authoring Effort and the Worksheet Generator

Building a lesson whose content was already prepared (video links, documents and activity instructions) took about 7 minutes from start to publish. This figure comes from 20 timed builds made while stress testing the platform, and it does not include creating the content itself. It shows that the authoring tool is unlikely to be the bottleneck, not that expert content is quick to produce. Where the lesson plan is already in a spreadsheet, it can be imported into the platform directly, which brings lesson creation to under 30 seconds. The block model also held up at a reasonable scale, with more than 110 lessons across eight classes (NCERT Grades 1 to 4 and NZC Years 1 to 4) in maths and science, which shows it can hold the structure of two different curricula and scale without problems.

The worksheet generator was tested on 15 lessons across both curricula and both subjects, with different mixes of sources: captioned videos, authored PDF and Word documents, and reading and activity text (Table A3, Appendix A). It produced 14 worksheets and correctly refused the one empty lesson without calling the AI. Across 158 questions there were no wrong numeric or science answers, regenerating gave a new draft each time, none of the 14 saved worksheets overwrote an existing block, and a worksheet took a median of 15.6 seconds and about $0.017 to produce. A video link it could not read was reported and the rest of the lesson was still used. Scores were highest for short captioned videos and clear documents. The test also showed where care is needed. Very long sources are cut at 60,000 characters, the prompt is not given the grade or curriculum, and some summaries add facts that are not in the source, which is why the admin's review of the summary matters. Every worksheet still needs an admin to review it before it is saved.

Many of the video links in the seeded lessons point to playlists or channel pages that the player cannot play, and need replacing with specific videos before delivery.

6.7 Validity and Limitations

The evidence in this chapter supports a modest claim: the platform implements the design it set out to, it handles two real curricula, its role checks, approval process and lesson-flow rules held in almost all of the tests that were run, and the person who reviewed it in detail understood its main ideas, such as the locked steps, without explanation. The tests also found some configuration and usability gaps, which are recorded for fixing. The evidence does not show that a facilitator with no subject expertise can deliver a lesson well, that students learn more, or that the platform is easy to use for people other than the few who have tried it. The central claim of the project, that any approved facilitator can deliver an expert-designed lesson, therefore needs further testing with real facilitators in a real classroom.

7. AI-Assisted Development

7.1 Tools and Pipeline

AI was used throughout the build, so this chapter records how, and which parts were my own decisions. Several tools were tried for different jobs before I settled on one pipeline (Table 8). Antigravity IDE, ChatGPT Codex and Claude Code were first tested as coding agents, and Claude Code was kept. ChatGPT (voice mode), Gemini and Claude were tried for refining ideas and requirements, and image and video tools for the lesson material and the poster, which sit outside the build pipeline.

Table 8. Tools tried for each job and what was kept in the pipeline.

Job	Tools tried	Kept in the pipeline
Writing code	Antigravity IDE, ChatGPT Codex, Claude Code	Claude Code
Refining ideas and requirements	ChatGPT (voice mode), Gemini, Claude	ChatGPT early on, then Claude alone from the mid-year report
Lesson material and the poster	Image and video tools	Not part of the build pipeline

The pipeline is shown in Figure 11. It began with ChatGPT, where I gave the project context and talked through ideas in voice mode to reach a first specification covering the problem, the goals, how the system should work, what each page needs and how it should look. I took that specification to Claude for a thorough rewrite, and then to Claude Code for a pass on whether it could actually be built, which produced one project brief. Around the time of my mid-year report I moved to Claude completely, so ChatGPT belongs to the early part of the project only.

From the brief, a session-by-session build plan broke the work into small pieces, each close to an agile sprint. Each session was built in Claude Code and tested by hand in the running app before the next began. The build started with Claude Sonnet 5 and moved to Claude Opus 5 and later Opus 5.5 as they became available, using "High" or "Extra" reasoning. Every three to four sessions, a full code review was run with Claude Opus 5.5 using Maximum reasoning, which I chose for finding bugs. Connectors, MCP servers and custom UI skills kept generated pages consistent with the design tokens, and context passed between sessions through structured handoff documents.

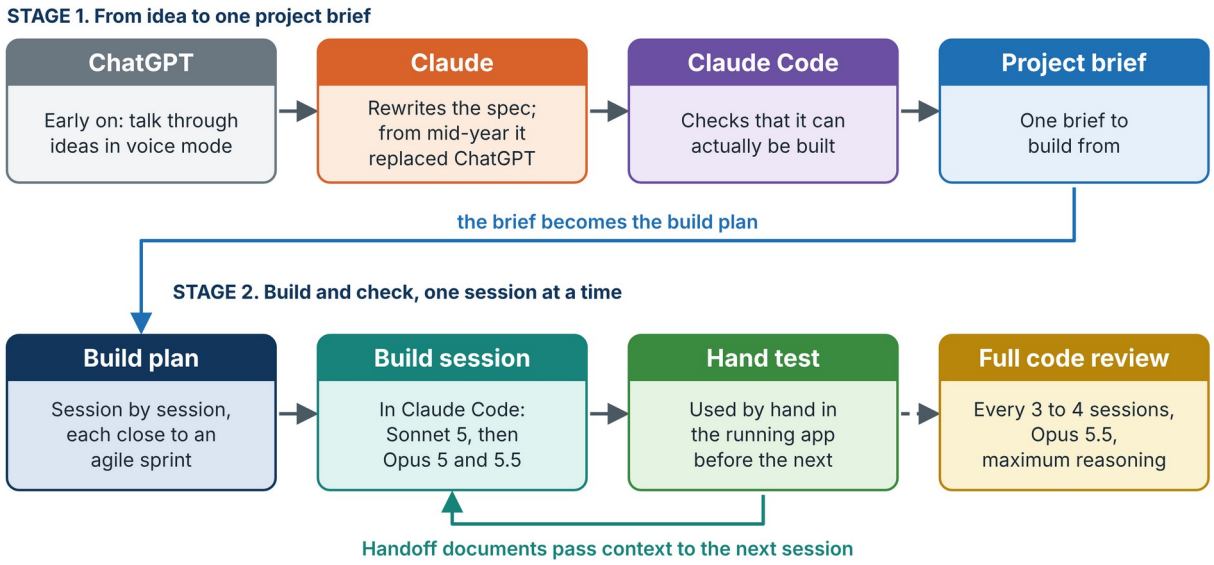


Figure 11. The tools and pipeline used to go from an idea to a tested build.

7.2 Guardrails

AI-written code can be fast and wrong, so the process had checks built in. Each feature began with a written specification, such as the one for the worksheet generator, which fixed what would be built before any code existed. The specification was updated as the code was written, so the record followed the system. Every session was tested by hand before the next, and the periodic review caught problems across sessions that testing one page at a time would miss. This did not catch everything. The problems found during the build included an invite-link bug caused by how the sign-in flow handled links, database connections running out, sessions dropping, buttons that could submit twice, and the gap where admin actions were protected only

in the interface (Section 5.5). Each was found by using the system or by review and not by the AI writing it, which is why the technical tests in Section 6.5 were run on the finished system.

7.3 What I Decided and What the AI Did

The decisions that define the platform were mine. These were the split between admins and facilitators, the five-step flow, approval before access, the locked first walkthrough, nine block types, the two approval steps of the worksheet generator, the choice to store no student data, and the direction of the admin redesign. I wrote and approved the specifications, tested each session, and decided what to keep, change or reject. The AI wrote most of the code, produced the first versions of designs, and proposed options, including the six interface styles in Section 5.7, from which I chose. I did not write most of the code line by line, so the contribution of this project is the design, the specification and the testing of the platform, with the AI as the main tool for implementation.

8. Discussion

8.1 Answering the Research Questions

The main question asked how a software platform can be designed so that any approved facilitator, regardless of subject expertise, delivers an expert-designed lesson as it was intended. The three sub-questions are taken first.

On authoring, an admin can capture a lesson's content, sequence and delivery instructions through typed blocks and facilitator notes, so the lesson can be run without the admin in the room. The evidence is that more than 110 lessons across two curricula were built this way, that building a prepared lesson takes about 7 minutes, and that the worksheet generator adds supplementary material with an admin approving each step (Sections 6.6 and 5.6). It does not show that the lessons are good. Quality still depends on the admin who writes them.

On delivery, the platform controls who delivers a lesson and guides them through it. Access is granted only after an admin approves a request, every action is checked on the server, and the walkthrough locks each block until the one before it is finished. The tests, the heuristic review and the tutor's feedback support this (Sections 6.3 to 6.5). What the platform controls is access and order. It cannot control how the facilitator then teaches.

On addressing the education gap, the benchmark shows that StudyBuddy combines decoupled delivery, structured guidance and adaptability in a way the other platforms do not. The learner insights support personalisation and engagement. The ratings are mine, from public descriptions, so this answer is a structured comparison and not a measurement.

Taken together, the project answers the main question as a design question. It shows one way to build a platform that enforces who delivers and in what order, and it shows that this can be built and populated with real curricula. Since real-world testing was out of the scope of this project, this platform does not have concrete evidence that facilitators without subject expertise deliver lessons as intended. However, I have designed and evaluated a platform that has the potential to make education more accessible and reliable.

8.2 Relation to the Literature

The literature identified structured guidance as one of the most effective ways to improve learning where teachers are scarce (Piper et al., 2018; Snilstveit et al., 2016; World Bank, 2018). StudyBuddy turns that idea into software, since the lesson is an ordered sequence of blocks and the facilitator is led through it step by step. Delivering a lesson as intended is called implementation fidelity (Carroll et al., 2007), and the walkthrough, the locked order and the server checks are mechanisms aimed at it. The project does not measure fidelity, so it shows the mechanisms and not their effect. The literature also found that video alone does not hold

learners, with low completion in self-paced courses (Jordan, 2015). The platform responds by placing a person in the room, and by mixing videos with activities, workbooks, discussions and breaks. Whether this holds attention depends on how the admin designs each lesson.

8.3 What Practitioners Challenged

The academic coordinator at my school supported the idea in several places, but she also raised points the platform does not yet answer. Her main challenge was that before any teaching comes managing a classroom, and the walkthrough prepares a facilitator for a lesson, not for a room of children. The platform assumes the facilitator can keep order, and the chat to the admins is the only support it offers during a lesson. This is the second of the two relationships that shaped the project (Section 2.3), between the facilitator and the students, and it is one reason the work was split and carried out in parallel. My part makes sure the lesson reaches the room as the admin designed it, and May's part (Thazin, 2026) aims to help the facilitator see how students are responding and feel better prepared to act on it. Insights alone do not tell a facilitator what to do next, so a further step would be to turn them into an actionable plan, such as suggested next steps for the lesson in progress. She also noted that lessons delivered by video must not become one-way, which puts the weight on the admin to build pauses, questions and activities into each lesson. She said that parents may resist new methods, which is an adoption question for a school and not something the software settles. Finally, she felt that very young children need their own teacher, which is part of why this work currently stays with Grades 1 to 4 and Years 1 to 4. These points are reflected in the future work.

8.4 How This Work Fits with May's

The project was split into two parts (Section 2.1). My part is the platform that lets an admin design a lesson and a facilitator deliver it. May's part is a tool that turns learner activity into insights for the facilitator (Thazin, 2026), which addresses what the benchmark shows to be weakest here, support during the lesson itself. The two parts are not yet joined in one product. Linking them, so that a facilitator is helped to respond to how students are doing, is the most natural next step and was suggested by my supervisor.

9. Conclusions and Future Work

9.1 Conclusions

Teaching quality varies from one classroom to the next, a lesson is often lost when its teacher leaves, and millions more teachers are needed worldwide. This project asked how a software platform could separate the people who design a lesson from the people who deliver it, so that an approved facilitator, whatever their subject expertise, can deliver an expert-designed lesson as it was intended. The answer built here is StudyBuddy, a web platform in which admins design lessons and approved facilitators deliver them. Admins build lessons from nine block types, supported by an AI worksheet generator whose output they approve twice. Facilitators gain access only through an admin's approval, walk through each lesson once with each block unlocking after the one before it, and can ask the admins a question at any point. Every action is checked on the server. The platform holds more than 110 lessons across the NCERT curriculum for Grades 1 to 4 and the New Zealand Curriculum for Years 1 to 4, in maths and science.

The evaluation shows that the platform does what its design sets out to do. Its role checks and lesson flow held in almost all of the tests, which also found some configuration and usability gaps, recorded for fixing; it handles two curricula; a prepared lesson takes about 7 minutes to build; and design feedback shaped several changes. The contribution of this project is the design of the platform, grounded in the literature, a working implementation, an evaluation with its limits stated, and an account of how AI was used to build it. Taken together, these show that the

design can put an expert's lesson in front of a class in the order and form the expert intended, a first step towards good teaching that depends less on who happens to be in the room.

9.2 Limitations

The platform has not been tested in a real classroom, so the central claim remains untested. The informal feedback came from a small number of people, mostly known to me, and the benchmark and heuristic review were my own work. The quality of every lesson depends on the admin who writes it, and the platform needs a screen and an internet connection. The project first planned a formal interview study under the University's human participants ethics process. On my supervisor's advice this was replaced by informal feedback, so what was gathered is design input and not research data.

9.3 Future Work

The first research priority is a classroom pilot with facilitators who are not subject experts, measuring how closely delivery follows the lesson design and what students learn. Several improvements follow from the feedback in this report. The walkthrough could include guidance on managing a classroom, which was the coordinator's main challenge. Admins could receive alerts when a facilitator falls behind or stops partway through a lesson. A facilitator code of conduct could be added to the approval step. Offline support would extend the platform to places without a reliable connection. The platform could be extended to younger and older age groups and other subjects, and linked to May's insights, which could personalise how a lesson is delivered and be turned into an actionable plan for the facilitator, as my supervisor suggested.

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References

Achuthan, K. (2025). Artificial intelligence and learner autonomy: A meta-analysis of self-regulated and self-directed learning. *Frontiers in Education, 10*, Article 1738751. https://doi.org/10.3389/feduc.2025.1738751

Banerjee, A., Banerji, R., Berry, J., Duflo, E., Kannan, H., Mukerji, S., Shotland, M., & Walton, M. (2017). From proof of concept to scalable policies: Challenges and solutions, with an application. *Journal of Economic Perspectives, 31*(4), 73-102. https://doi.org/10.1257/jep.31.4.73

Carroll, C., Patterson, M., Wood, S., Booth, A., Rick, J., & Balain, S. (2007). A conceptual framework for implementation fidelity. *Implementation Science, 2*, Article 40. https://doi.org/10.1186/1748-5908-2-40

Chen, C.-M. (2009). Personalized e-learning system with self-regulated learning assisted mechanisms for promoting learning performance. *Expert Systems with Applications, 36*(5), 8816-8829. https://doi.org/10.1016/j.eswa.2008.11.026

Coates, H., James, R., & Baldwin, G. (2005). A critical examination of the effects of learning management systems on university teaching and learning. *Tertiary Education and Management, 11*(1), 19-36. https://doi.org/10.1080/13583883.2005.9967137

Daga-As, C. P., & Guntalidad, J. A. (2025). Breaking boundaries: The real lives of alternative learning system facilitators. *Journal of Educational Management and Strategy, 4*(2), 133-142. https://doi.org/10.57255/jemast.v4i2.1518

Darling-Hammond, L. (2000). Teacher quality and student achievement. *Education Policy Analysis Archives, 8*(1), Article 1. https://doi.org/10.14507/epaa.v8n1.2000

Garrison, D. R., Anderson, T., & Archer, W. (1999). Critical inquiry in a text-based environment: Computer conferencing in higher education. *The Internet and Higher Education, 2*(2-3), 87-105. https://doi.org/10.1016/S1096-7516(00)00016-6

Global Education Monitoring Report Team. (2023). *Can countries afford their national SDG 4 benchmarks?* UNESCO. https://unesdoc.unesco.org/ark:/48223/pf0000385004

Hooshyar, D., Weng, X., Sillat, P. J., Tammets, K., Wang, M., & Hamalainen, R. (2024). The effectiveness of personalized technology-enhanced learning in higher education: A meta-analysis with association rule mining. *Computers & Education, 223*, Article 105169. https://doi.org/10.1016/j.compedu.2024.105169

Huang, R., Yin, Y., Zhou, N., & Lang, F. (2026). Artificial intelligence in K-12 education: An umbrella review. *Computers and Education: Artificial Intelligence, 10*, Article 100519. https://doi.org/10.1016/j.caeai.2025.100519

Islam, S., & Chowdhury, Z. F. (2019, October 13). *Educating a generation: Bangladesh's barefoot teachers*. BRAC. https://blog.brac.net/educating-a-generation-bangladeshs-barefoot-educators/

Jordan, K. (2015). Massive open online course completion rates revisited: Assessment, length and attrition. *The International Review of Research in Open and Distributed Learning, 16*(3), 341-358. https://doi.org/10.19173/irrodl.v16i3.2112

Kim, R., Olfman, L., Ryan, T., & Eryilmaz, E. (2014). Leveraging a personalized system to improve self-directed learning in online educational environments. *Computers & Education, 70*, 150-160. https://doi.org/10.1016/j.compedu.2013.08.006

Kizilcec, R. F., Piech, C., & Schneider, E. (2013). Deconstructing disengagement: Analyzing learner subpopulations in massive open online courses. In *Proceedings of the Third*

International Conference on Learning Analytics and Knowledge (pp. 170-179). ACM. https://doi.org/10.1145/2460296.2460330

Means, B., Toyama, Y., Murphy, R., Bakia, M., & Jones, K. (2010). *Evaluation of evidence-based practices in online learning: A meta-analysis and review of online learning studies*. U.S. Department of Education.

Nielsen, J. (1994). *10 usability heuristics for user interface design*. Nielsen Norman Group. https://www.nngroup.com/articles/ten-usability-heuristics/

Piper, B., DeStefano, J., Kinyanjui, E. M., & Ong'ele, S. (2018). Scaling up successfully: Lessons from Kenya's Tusome national literacy program. *Journal of Educational Change, 19*(3), 293-321. https://doi.org/10.1007/s10833-018-9325-4

Snilstveit, B., Stevenson, J., Menon, R., Phillips, D., Gallagher, E., Geleen, M., Jobse, H., Schmidt, T., & Jimenez, E. (2016). *The impact of education programmes on learning and school participation in low- and middle-income countries: A systematic review summary report*. International Initiative for Impact Evaluation (3ie).

Thazin, M. (2026). *Making Education More Accessible & Reliable: AI-Driven Facilitator-Based Teaching* [Final report]. Department of Electrical, Computer, and Software Engineering, The University of Auckland.

United Nations Educational, Scientific and Cultural Organization. (2015). *Rethinking education: Towards a global common good?* UNESCO.

United Nations Educational, Scientific and Cultural Organization. (2020). *Global education monitoring report 2020: Inclusion and education: All means all*. UNESCO. https://unesdoc.unesco.org/ark:/48223/pf0000373718

United Nations Educational, Scientific and Cultural Organization. (2024). *Global report on teachers: Addressing teacher shortages and transforming the profession*. UNESCO. https://www.unesco.org/en/articles/global-report-teachers-addressing-teacher-shortages-and-transforming-profession

Veine, S., Anderson, M. K., Skancke, L. B., & Wallin, P. (2022). Educating learning assistants as facilitators: Design challenges and experiences of practice. *Journal of Experiential Education, 46*(4), 491-512. https://doi.org/10.1177/10538259221147010

World Bank. (2018). *World development report 2018: Learning to realize education's promise*. World Bank.

World Bank. (2022). *The state of global learning poverty: 2022 update*. World Bank.

World Bank. (2025). *Learning poverty data*. https://data360.worldbank.org/en/dataset/WB_LPGD

Appendices

Appendix A: Supporting Tables

The tables below support Chapter 6. Each is summarised in the text where it is cited.

Table A1. Nielsen's ten heuristics and how StudyBuddy responds to each.

Heuristic	How the platform responds
1. Visibility of system status	The step list marks every block as done, current or locked, with an "X of Y complete" badge that updates as steps are finished
2. Match with the real world	Class cards follow the Canvas course-card pattern most users already know
3. User control and freedom	Deleting a block needs a second click within three seconds, which prevents accidental loss without a pop-up
4. Consistency and standards	The three themes are built from one component set and one token system
5. Error prevention	Changing a password in Settings requires the current password, and a sign-up request stays a request, with no account, until an admin approves it
6. Recognition rather than recall	Every nested page shows a breadcrumb
7. Flexibility and efficiency of use	Create lesson from inside a subject pre-fills the class and subject
8. Aesthetic and minimalist design	The admin home page was redesigned around the actions an admin needs next
9. Recognise and recover from errors	The worksheet generator reports a source it cannot read and still completes the rest
10. Help and documentation	The settings menu links to the design record and the worksheet generator specification

Table A2. Response times on a local production build (25 runs each, in milliseconds).

Page or action	Median	p95
Admin dashboard	757	966
Facilitator dashboard	829	931
Lessons index (125 lessons)	556	1,214
Lesson player, 5 steps	1,149	1,231
Lesson player, 25 steps	1,138	1,567
Add block	1,124	1,239
Mark step done	1,123	1,185

Table A3. Worksheet generator test set.

Measure	Result
Lessons tested	15 (14 worksheets; 1 empty lesson refused, no AI call)
Questions checked	158; no wrong numeric or science answers
Summary accuracy (0 to 2)	Mean 1.64
Curriculum fit (0 to 2)	Mean 1.29
Answer key (0 to 2)	Mean 1.71
Edits needed per worksheet	Median 2
Regenerate	Worked in 2 of 2
Existing blocks overwritten	0 of 14
Time and cost per worksheet (20 runs)	Median 15.6 s and $0.017